

Meeting 1203

黃丞暘

國立東華大學應數統計所

December 03, 2025

Index

1 11.26 Discussion

2 11.26 Todo

3 Result - What I do this time

11.26 Discussion

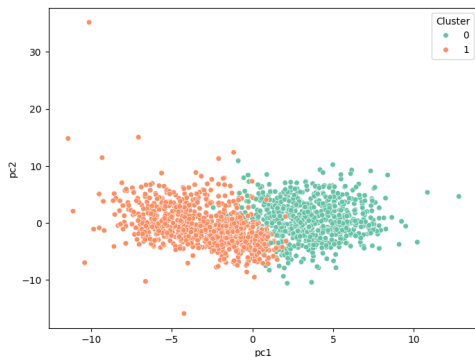
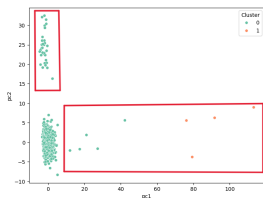
Semiconductor Manufacturing Process

- ▶ 抽掉 outlier 讓 PCA 更散更清楚
- ▶ (Y_i, X_i) iid $F(\cdot)$

11.26 Todo

- ▶ 繼續分成幾群再套模型比較最合適的
- ▶ 對 outlier 做質性的統計方法
- ▶ 在降維分群之前做 SMOTE 看看效果

Results I PCA + KMeans, 2 clusters



cluster	0	1
$y = 0$	660	770
$y = 1$	58	37

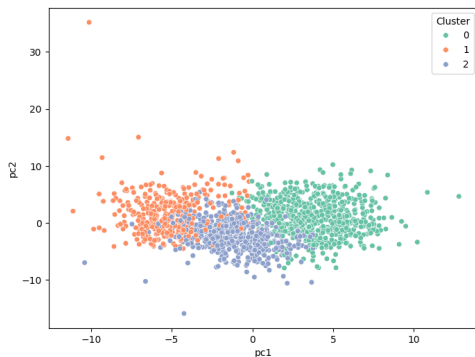
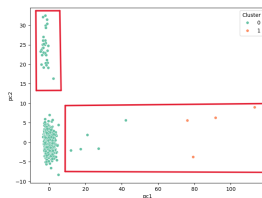
Results I

ns=10, 5-folds cv - mean

model	ACC	FAR	ROCAU
LR	0.8520	0.1011	0.5459
SVM	0.9304	0.0041	0.5027
DT	0.8149	0.1415	0.5309
RF	0.9304	0.0048	0.5074
XGB	0.9272	0.0096	0.5145

model	C_0 ACC	C_0 FAR	C_0 ROCAU	C_1 ACC	C_1 FAR	C_1 ROCAU
LR	0.8315	0.1152	*0.5545	*0.9009	*0.0649	*0.5586
SVM	0.9178	0.0015	0.4992	*0.9542	0.0000	0.5000
DT	*0.8273	0.1258	*0.5848	*0.8934	*0.0753	*0.5802
RF	0.9151	0.0045	0.4977	*0.9554	0.0000	*0.5143
XGB	0.9081	0.0197	*0.5341	*0.9554	*0.0013	*0.5261

Results II PCA + KMeans, 3 clusters



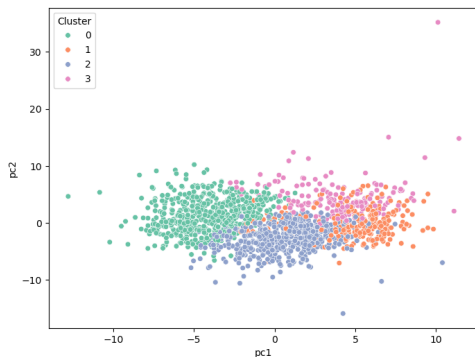
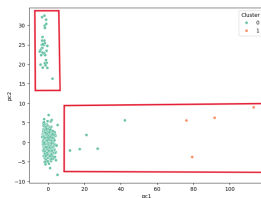
cluster	0	1	2
$y = 0$	592	327	511
$y = 1$	49	19	27

Results II ns=10, 5-folds cv - mean

model	C_0 ACC	C_0 FAR	C_0 ROCAU	C_1 ACC	C_1 FAR	C_1 ROCAU
LR	0.8268	0.1149	0.5036	*0.8930	*0.0612	0.5278
SVM	0.9251	0.0000	*0.5111	*0.9422	*0.0031	0.4985
DT	*0.8315	*0.1200	*0.5633	*0.8701	*0.0856	0.5072
RF	0.9220	0.0017	0.4992	*0.9451	0.0000	0.5000
XGB	0.9111	0.0169	0.5116	*0.9335	0.0122	0.4939

model	C_2 ACC	C_2 FAR	C_2 ROCAU
LR	*0.9183	*0.0470	*0.6032
SVM	*0.9498	0.0000	0.5000
DT	*0.8942	*0.0705	*0.5747
RF	*0.9498	0.0000	0.5000
XGB	*0.9386	0.0137	0.5098

Results III PCA + KMeans, 4 clusters



cluster	0	1	2	3
$y = 0$	513	231	523	163
$y = 1$	39	12	32	12

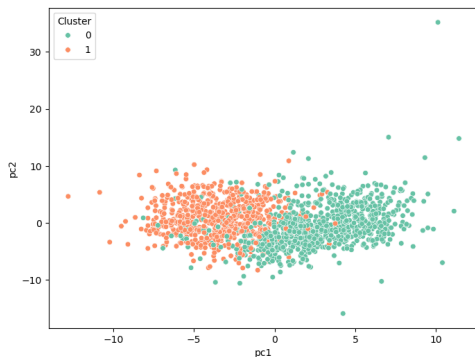
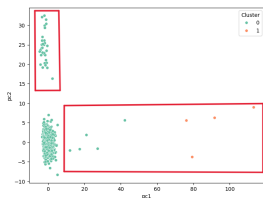
Results III

ns=3, 5-folds cv - mean

model	C_0 ACC	C_0 FAR	C_0 ROCAU	C_1 ACC	C_1 FAR	C_1 ROCAU
LR	*0.8532	*0.0937	0.5335	*0.9095	*0.0476	0.5095
SVM	0.9294	0.0000	0.5000	*0.9507	0.0000	0.5000
DT	*0.8224	*0.1228	0.4886	*0.8849	*0.0821	*0.5923
RF	0.9294	0.0000	0.5000	*0.9507	0.0000	0.5000
XGB	0.9203	0.0136	*0.5182	*0.9466	*0.0043	0.4978

model	C_2 ACC	C_2 FAR	C_2 ROCAU	C_3 ACC	C_3 FAR	C_3 ROCAU
LR	*0.8973	*0.0631	*0.5947	*0.8857	*0.0803	*0.6765
SVM	*0.9423	0.0000	0.5000	*0.9314	0.0000	0.5000
DT	*0.8468	*0.1069	0.4918	*0.8514	*0.0860	0.4570
RF	*0.9423	0.0000	0.5000	*0.9314	0.0000	0.5000
XGB	*0.9423	*0.0019	0.5133	0.9200	0.0123	0.4938

Results IV PCA + AgglomerativeC, 2 clusters



cluster	0	1
$y = 0$	822	608
$y = 1$	37	58

Results IV ns=10, 5-folds cv - mean

model	C_0 ACC	C_0 FAR	C_0 ROCAU	C_1 ACC	C_1 FAR	C_1 ROCAU
LR	*0.8906	*0.0766	0.5403	0.8259	0.1200	*0.5688
SVM	*0.9569	0.0000	0.5000	0.9129	*0.0016	*0.5075
DT	*0.8917	*0.0766	*0.5563	0.8048	*0.1383	*0.5339
RF	*0.9581	0.0000	*0.5125	0.9099	*0.0033	0.4984
XGB	*0.9593	0.0136	*0.5268	0.9024	0.0214	*0.5401

	Pass	Fail
All	0.937705	0.062295
Outlier	0.928571	0.071429

Feature	mean _{all}	mean _{out}	Z _{mean,out}
x80	0.002938	0.028386	1.128814
x60	2.592655	10.330188	0.968735
x104	-0.009867	-0.007076	0.917956
x39	86.820432	87.185567	0.883545
x511	54.871962	85.931150	0.846847
x76	-0.006494	-0.023774	-0.791008
x56	2855.748253	2874.928571	0.748401
x29	69.610802	67.144440	-0.726280
x274	20.157059	22.708264	0.671723
x412	2.612735	2.932202	0.580779

The End

Questions & Comments